Group No:6
Project Title
MUX and Decoder Simulator

Example 1: 4-to-1 Multiplexer (MUX)

Given

$D_0 = 1$ $D_1 = 0$ $D_2 = 1$ $D_3 = 0$

$S_1 = 0$ $S_0 = 0$

Solution

The select lines are $S_1S_0 = 00$
Therefore, D_0 is selected

Output:

$Y = D_0 = 1$

Final answer: $Y = 1$

Example 2: 2-to-4 Decoder

Given

$A = 1$ $B = 0$

Solution

The binary input is $AB = 10$. Therefore, output Y_2 becomes HIGH(1), and all other output remain LOW(0)

Output

$Y_0 = 0$
$Y_1 = 0$
$Y_2 = 1$
$Y_3 = 0$

Final answer: $Y_2 = 1$

Example.

Given:

$D_0 = 0$ $D_1 = 1$ $D_2 = 0$ $D_3 = 1$

$S_1 = 1$ $S_0 = 0$

Solution:

The select lines are $S_1 S_0 = 10$ Therefore, D_2 is selected

Output:

$Y = D_2 = 0$

Final Answer: $Y = 0$